

# COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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## ABSTRACT

Diffusion probabilistic models (DPMs) have demonstrated remarkable progress in generative tasks, such as image and video synthesis, yet they still often produce hallucinated samples (*hallucinations*) that conflict with real-world knowledge, such as generating an implausible duplicate cup floating beside another cup. The core obstacle we target is that, despite their prevalence, hallucinations lack any feasible methodology for systematic quantification, which stalls progress on the problem and obscures pathways toward next-generation generative models that respect factual constraints. This paper makes four contributions toward closing that gap. *First*, we isolate and formally define a specific, measurable form of hallucination, which we term **counting hallucination**: the generation of an incorrect number of instances or structured objects, such as a hand image with six fingers, despite such patterns being absent from the training data. *Second*, we construct **CountHalluSet**, a dataset suite with well-defined counting criteria comprising ToyShape, SimObject, and RealHand, and pair it with a standardized, reproducible evaluation protocol for quantifying counting hallucinations. *Third*, using this protocol we deliver the first systematic study of how DPM sampling conditions, including solver type, ODE solver order, sampling steps, and initial noise, affect counting hallucination levels. *Fourth*, we analyze the correlation between counting hallucinations and common evaluation metrics such as the Fréchet Inception Distance (FID), and show that widely-used perceptual metrics fail to capture counting hallucinations consistently. Together, these contributions establish counting hallucination as a tractable entry point for measuring and ultimately mitigating factual errors in diffusion models.

## 1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion model, have achieved considerable success in generative modeling, producing high-quality samples across various tasks such as text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Despite these impressive advancements, diffusion models still struggle to faithfully capture the factual properties of their training data, and often generate outputs that violate consistency with real-world knowledge. The failure we focus on throughout this paper is the generation of objects with an incorrect number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Recent studies have increasingly focused on identifying and analyzing common failure modes in text-guided diffusion models, and they help position the gap this paper fills. For instance, Borji (2023) systematically catalogued several qualitative shortcomings in text-to-image models. Additionally, Liu et al. (2024) proposed an adversarial search method to uncover undesirable behaviors and failure cases in text-to-image generation, while Wu et al. (2023) employed an iterative approach to discover and mitigate spurious correlations. What these efforts share, and what we address, is that they surface failures qualitatively without a protocol that quantifies them.

More recently, *hallucination*, a specific failure mode in diffusion models, has garnered attention in the research community (Aithal et al., 2024). Unlike previously discussed text-prompt-based failure modes, which can be evaluated by measuring the alignment between the input conditions and outputs (e.g., CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)),

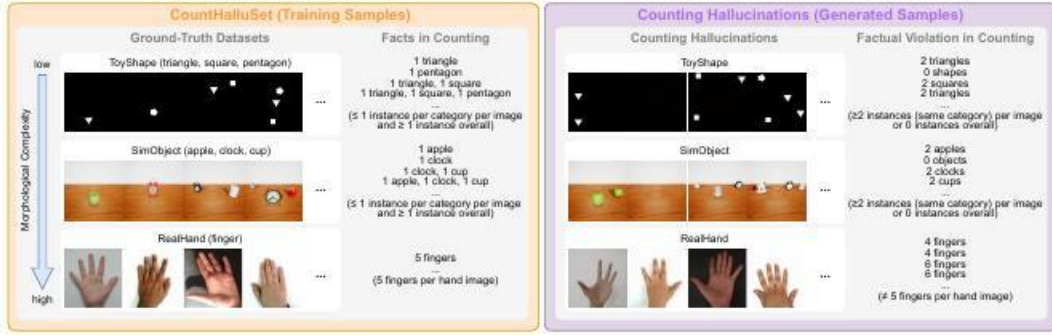


Figure 1: Example comparison between the proposed **CountHalluSet** used for training (left) and the corresponding **counting hallucinations** generated by the model (right). Each dataset contains different objects with specific counting criteria. For example, in ToyShape and SimObject, each image contains at most one instance per category and at least one instance overall, whereas in RealHand, each image contains exactly five fingers. Counting hallucinations refer to generated images that violate the dataset’s counting rules, such as two apples in an image or an image containing no objects but a normal background, even though these patterns never appear in the SimObject dataset.

or more obvious failure modes such as severe distortions, blur, and artifacts, hallucinated samples (*hallucinations*) often appear visually plausible, and emerge independently of the conditioning signal, manifesting as subtle violations of factual consistency with respect to the training data, such as incorrect object counts, geometries that appear plausible but are physically impossible, or other breaches of physical laws. These fine-grained distinctions between valid samples and hallucinations present a critical challenge for current AI systems, which often fail to detect them reliably. For example, even state-of-the-art large vision-language models like GPT-4o still struggle with basic tasks like accurately counting the number of fingers in a clear hand image. This difficulty motivates the central need this paper answers: a standardized, well-defined protocol for reliably quantifying hallucinated samples. We argue such a protocol is essential because it provides the foundation for designing next-generation generative models with strict factual consistency, an ability that is critical for their role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the various common types of hallucinated elements, such as counts, colors, spatial relationship, geometry, and physical realism, counting errors are particularly suitable for systematic study because they are not only easy to identify but also straightforward to quantify. This observation grounds our *first contribution*: we define a specific, measurable form of hallucination termed **counting hallucination**. In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values. For example, an image depicting a hand with six fingers is classified as a counting hallucination, since such a pattern never occurs in the RealHand dataset, as shown in the last row of Fig. 1. As our *second contribution*, and to make the study of such phenomena reproducible, we construct the dataset suite CountHalluSet, consisting of three datasets that span a spectrum of morphological complexity for the countable objects: ToyShape (*triangle, square, pentagon*), SimObject (*mug, apple, clock*), and RealHand (*finger*). Figure 1 illustrates representative examples from these datasets alongside corresponding counting hallucinations.

The critical step in this protocol is the accurate identification and quantification of objects, which we accomplish with counting models tailored to each dataset. Building on these datasets and counting models, our *third contribution* is a systematic empirical study of how various denoising conditions (e.g., sampling steps, initial noise, the order of ODE solvers, and solver type) within diffusion models affect counting hallucination rates. Our *fourth contribution* is a correlation analysis between the counting hallucination rate and perceptual metrics, including the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Concretely, these contributions surface the following findings, which we return to and support throughout the paper:

- **Numerical strategies do not transfer to real objects.** Commonly applied numerical strategies in diffusion models, such as increasing sampling steps from 25 to 100, often fail

to mitigate counting hallucinations when generating human hands, even though these same techniques are effective on simpler synthetic datasets.

- **Perceptual metrics miss counting hallucinations.** We observe only a weak correlation between the counting hallucination rate and perceptual metrics, whereas a strong correlation emerges between the non-counting failure rate and these same metrics.
- **A mitigation paradigm follows directly from our analysis.** Building on the finding above, we propose a novel and promising training paradigm, termed joint-diffusion models, which significantly reduces both counting hallucinations and non-counting failures on the RealHand dataset.

## 2 RELATED WORK

### 2.1 HALLUCINATIONS IN DIFFUSION MODELS

Inspired by the growing attention to hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), the study of hallucinations in unconditional diffusion models has also garnered increasing interest. In particular, Aithal et al. (2024) proposed a qualitative hypothesis suggesting that hallucinated samples may arise from data interpolating between distinct modes. Despite this insight into potential causes, a systematic *quantitative* analysis of hallucinations in diffusion models remains lacking, and it is precisely this missing quantitative footing that our protocol and datasets supply.

This gap directly motivates our second and third contributions: accurately identifying and quantifying hallucinations in diffusion-generated outputs. Such an investigation deepens the understanding of undesirable behaviors exhibited by these models, and, as our experiments make explicit, clarifies how the magnitude of denoising errors, such as variations in sampling steps or the order of ODE solvers, affects the severity of hallucinations.

### 2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are two widely used metrics for evaluating the performance of text-guided generated images. However, such evaluation methods that rely on language-based models are inherently limited when applied to assess hallucinated images. One line of evidence is the statistical lower bound on the propensity of pretrained language models to hallucinate certain fact types, which constrains the upper bound of detection accuracy based solely on language models (Kalai & Vempala, 2024). Another indication is the persistent performance gap between human annotators and LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

In addition, several widely used quantitative evaluation metrics, such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021), are commonly employed in image generation tasks, all relying on feature embeddings extracted from a pretrained Inception-v3 model (Szegedy et al., 2016). The open question these metrics leave unanswered, *whether they can capture the severity of hallucinations in generated datasets*, is exactly the question our fourth contribution takes up. We address this gap from the perspective of counting hallucinations, and our correlation analysis in Sec. 5.3 shows that the answer is negative.

## 3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

### 3.1 FORWARD (DIFFUSION) PROCESS

This section fixes the notation and sampling machinery that our evaluation protocol and findings build on. **Forward Process.** Given the observed data  $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$ , Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) introduce a *forward process* (a.k.a, *diffusion process*) that gradually perturbs the data by adding Gaussian noise over  $T$  discrete timesteps. At each timestep  $t$ , the marginal distribution of the noisy sample  $\mathbf{x}_t$  conditioned on the original data  $\mathbf{x}_0$

follows:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where  $t \in \{0, 1, \dots, T\}$  and  $\{\beta_i\}_{i=1}^T$  is a pre-defined noise schedule dependent on  $t$ , commonly chosen to be a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The factor  $\bar{\alpha}_t$  controls the signal-to-noise ratio at step  $t$ :  $\bar{\alpha}_0 = 1$  (no noise) and  $\bar{\alpha}_T \approx 0$  (nearly pure noise). An equivalent formulation of Eq. (1) is:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As  $T$  becomes sufficiently large, the marginal distribution  $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$  approaches a standard Gaussian distribution  $\mathcal{N}(\mathbf{0}, \mathbf{I})$ . In widely-used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021),  $T$  is typically set to 1000.

### 3.2 REVERSE (DENOISING) PROCESS.

**Training Objective.** The *reverse process* (a.k.a., *denoising process*) of DPMs iteratively recovers the observed data starting from the diffused data  $\mathbf{x}_T$ . Specifically, DPMs predict the noise  $\boldsymbol{\epsilon}$  from the data  $\mathbf{x}_t$  using a neural network  $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ , known as the *noise prediction model*. The parameter  $\theta$  is optimized by minimizing the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\pi(t) \|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2], \quad (3)$$

where  $\pi(t) > 0$  is a weighting function.

**Ancestral Sampling.** Given noisy data  $\mathbf{x}_T$ , samples from the original distribution  $q_0(\mathbf{x}_0)$  are obtained by reversing the diffusion process with the same number of steps, a procedure commonly referred to as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where  $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$  is the only unknown term, referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function  $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$  via the noise prediction model, so that an equivalent formulation of Eq. (3) is:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \quad (5)$$

**ODE-Based Accelerated Sampling.** The ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992); it has an associated deterministic process that can be formulated as an ordinary differential equation (ODE) (a.k.a., the *probability flow ODE* (Song et al., 2022)). Because solver type and order are among the sampling conditions our study varies, this deterministic view is central to the experiments that follow:

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau) \mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2} \beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2} \beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where  $\tau \in [T, 0]$ . Let  $\mathbf{x}(\tau)$  denote the ideal trajectory that satisfies this ODE with the true score function  $h(\mathbf{x}(\tau), \tau)$  and starts from an ideal diffused data  $\mathbf{x}(T)$  drawn from the true diffused distribution  $q_T(\mathbf{x}(T))$ . Set the discrete time steps as  $\tau_k = T - k\Delta t$  for  $k = 0, 1, \dots, N$ , where  $\tau_N = 0$ . Applying the variation-of-constants formula for the exact solver over one step:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where  $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$  is the state transition operator for the linear part  $\frac{dz}{d\tau} = f(\tau)z$ . Here  $\Delta t > 0$  denotes the step size, allowing ODE-based solvers to accelerate sampling with larger step sizes than ancestral sampling, and the total number of sampling steps is  $T/\Delta t$ . Since the linear part  $\mathcal{G}(t_2, t_1)$  can be computed in closed form, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) devote their numerical effort, such as higher solver orders and more sampling steps, to accurately approximating the nonlinear integral term involving  $h$ . These are exactly the knobs whose effect on counting hallucinations we quantify in Sec. 5.2; the supporting theoretical analysis can be found in Appendix A.1 to A.3.

## 4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section develops the first two contributions of the paper: the formal definition of counting hallucination and the CountHalluSet suite with its standardized evaluation protocol. We first define counting criteria and the associated counting hallucinations, then introduce the three datasets in CountHalluSet. For each dataset, we specify its counting criteria and corresponding counting hallucinations, and outline the standardized protocol used to quantify them, so that the empirical study in Sec. 5 rests on a reproducible measurement pipeline.

### 4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

The foundation of our first contribution is a definition that is general enough to span all three datasets yet precise enough to be computed automatically. We first state the counting criteria that a training dataset imposes, then define counting hallucinations as deviations from these predefined constraints.

**Counting criteria in a training dataset.** Let  $\mathcal{D}_{\text{ref}}$  be a training dataset. For any reference sample  $\mathbf{x} \in \mathcal{D}_{\text{ref}}$ , the number of objects of category  $c$  in  $\mathbf{x}$ ,  $N_c(\mathbf{x})$ , are given by a predefined set  $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$  (e.g.,  $\{0, 1\}$  or  $\{1, 2, 3\}$ ), i.e.,  $N_c(\mathbf{x}) \in \mathcal{S}_c$ . We further require that a valid reference sample contains at least one object, i.e.  $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$ , where  $C$  is the set of considered categories.

**Counting hallucinations as deviations from the counting criteria.** Given a generated sample  $\hat{\mathbf{x}}$ , we say  $\hat{\mathbf{x}}$  exhibits a *counting hallucination* with respect to  $\mathcal{D}_{\text{ref}}$  if both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image  $\hat{\mathbf{x}}$  should exhibit sufficient visual quality such that the background and any present objects are clearly discernible. This ensures that counting hallucinations can be distinguished from other prominent failure modes, such as severe degradations. We define a binary indicator  $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$ , referred to as the *counting-ready indicator*, where 1 denotes that the image quality is sufficient for counting, and 0 indicates otherwise. In practice this indicator can be determined via human annotation, by a classifier calibrated against human annotations, or by other automated criteria.
2. **Violation of counting facts.** A violation of counting facts occurs if there exists a category  $c$  such that  $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ , or if the image contains no objects (i.e.,  $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$ ) as the dataset requires at least one object per sample. In practice, we use a counting model to predict object counts in generated images as evidence for such violations.

A counting hallucination (CH) is identified if its quality is sufficient for counting ( $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$ ) and the predicted counts deviate from the counting criteria ( $\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$  or  $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$ ). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left( \exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

### 4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

Realizing our second contribution, the CountHalluSet suite consists of three datasets that deliberately span a spectrum of morphological complexity for studying counting hallucinations: ToyShape, SimObject, and RealHand. This progression from synthetic shapes to real hands is what later lets us isolate how object complexity drives hallucination rates. Examples from each dataset and corresponding counting hallucinations are shown in Fig. A1 and A2, with additional statistical summaries in Appendix A.4.

#### 4.2.1 TOYSHAPE

**Dataset description.** As the simplest end of the complexity spectrum, the ToyShape dataset comprises 30,000 images with three geometric shapes: *triangle*, *square*, *pentagon*. All shapes are white, with an equal area of 120 pixels, and no shapes overlap within any image. The counting criteria specify that each image contains at most one instance of each shape category and at least one shape in total, as shown in the top-left of Fig. 1. Formally, we have  $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$ . For each class  $c \in C$ , we set  $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ , subject to the constraint  $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$ .

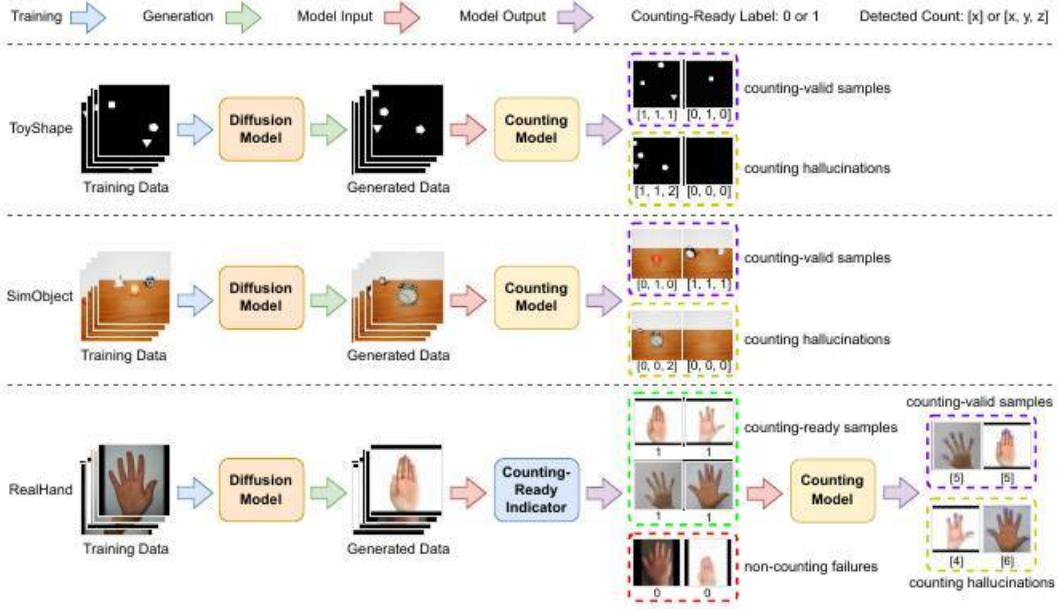


Figure 2: The standardized evaluation protocol of the CountHalluSet suite, the core of our second contribution. For ToyShape and SimObject, generated images are assessed for counting directly. For RealHand, a counting-ready indicator (a binary classifier aligned with human judgments) is added to first separate counting hallucinations from other non-counting failures, such as severely deformed fingers. Count labels correspond to ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

**Counting hallucinations.** Applying the criteria above, any image generated by a model trained on ToyShape that contains two or more shapes of the same category is a counting hallucination, such as the image containing two pentagons in the top-right region of Fig. 1. Empty images (i.e., containing only the black background without any shapes) are likewise classified as counting hallucinations. Further examples are provided in Appendix A.5.

**Counting model used in ToyShape.** To identify counting errors we use a counting model, and forgo a counting-ready indicator here because the dataset’s simplicity makes severely degraded images rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape images, with each category appearing 0–3 times per sample (i.e.,  $S_c = \{0, 1, 2, 3\}$ ), so as to cover a wide range of plausible scenarios spanning counting-valid samples and counting hallucinations and thereby enable accurate quantification. This model reaches over 99.9% counting accuracy on generated data, giving us high confidence in the measured hallucination rates.

**Quantification of counting hallucinations.** The protocol proceeds in three steps, as illustrated in Fig. 2. Given a diffusion model trained on ToyShape, we (i) generate a set of samples equal in size to the training dataset (30,000 images), (ii) apply the counting model to predict the number of objects in each generated image, and (iii) flag every image that violates the counting criteria as a counting hallucination.

#### 4.2.2 SIMOBJECT

**Dataset description.** Occupying the middle of the complexity spectrum, the SimObject dataset includes 30,000 rendered images of everyday objects. It comprises three object categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations. All objects are randomly placed on a wooden tabletop under fixed lighting conditions, with minimal mutual occlusion if any, as shown in the middle of the left side of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), featuring photorealistic rendering with accurate simulation of object geometry, material attributes, lighting, and reflections. The counting criteria for this dataset mirror those of ToyShape (see Sec. 4.2.1). Formally, we have  $C = \{mug, apple, clock\}$ . For each class  $c \in C$ , we set  $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ , subject to the constraint  $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$ .

**Quantification of counting hallucinations.** Because the counting criteria match, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol all follow the same three-step procedure used for ToyShape, as shown in Fig. 2.

#### 4.2.3 REALHAND

**Dataset description.** At the most complex end of the spectrum, and to investigate counting hallucinations in real-world scenarios, we construct the RealHand dataset comprising 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle<sup>1</sup>, and Roboflow<sup>2</sup>. Specifically, we aggregate raw images from these sources and remove those with blur or artifacts. Each image depicts a human hand with five fingers, capturing both dorsal (back) and palmar (front) views under varying lighting conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria specify that each image contains exactly five fingers, i.e.,  $C = \{\text{finger}\}$  with  $N_c(\mathbf{x}) \in S_c = 5$  for each  $c \in C$ , and  $\sum_{c \in C} N_c(\mathbf{x}) = 5$ .

**Counting hallucinations and non-counting failures.** Generating realistic hand images is inherently prone to unexpected failures beyond counting errors such as extra or missing fingers, including blurriness, severely deformed fingers, and visual artifacts. Distinguishing these two kinds of failure is precisely what makes RealHand the most demanding case, and we handle it with a counting-ready indicator (i.e., a binary classifier). Examples of counting hallucinations and non-counting failure samples are provided in Appendix A.7.

**Counting-ready indicator and counting model used in RealHand.** To separate counting hallucinations from the failure modes above, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as a counting-ready indicator aligned with human judgments, and we finetune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images for counting. Judged against human evaluation of 500 randomly sampled images by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively, which underpins the reliability of the RealHand measurements reported later. Examples of YOLO detection results are provided in Appendix A.6.

**Quantification of counting hallucinations.** Unlike ToyShape and SimObject, where all generated images are evaluated directly because severe failure cases (e.g., distorted or unrecognizable objects) are rare, RealHand requires an additional filtering step. Concretely, we first apply the counting-ready indicator to retain hand images with sufficient visual quality, and only then use the counting model to predict the number of fingers, as shown in Figure 2.

## 5 EXPERIMENTS

### 5.1 IMPLEMENTATION DETAILS

**Training of Diffusion Models.** For the ToyShape dataset we train a DDPM (Ho et al., 2020) from scratch, and for the SimObject and RealHand datasets we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on the CelebA-HQ dataset (Karras et al., 2018). The time step  $T$  is set to 1,000 with a standard linear noise schedule. Images are resized to  $128 \times 128$  for DDPMs and  $256 \times 256$  for LDMs. We optimize with Adam (Kingma, 2015) at a learning rate of 0.0001, running for 150k, 300k, and 80k steps on the ToyShape, SimObject, and RealHand datasets, respectively, with an effective batch size of 256.

**Inference of Diffusion Models.** For ancestral sampling (i.e., DDPM) we use 1,000 sampling steps. For the ODE-based solvers, we deliberately span solver order by evaluating both a first-order method, DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)), and a second-order method, DPM-Solver-2 (Lu et al., 2022). Each solver is evaluated under three widely-used step configurations, 25, 50, and 100 steps, so that our study covers solver type, solver order, and step count together. For the RealHand dataset, the counting model employs a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless noted otherwise, the number of generated samples matches the number of training samples in each evaluation, and all results are averaged over three random seeds.

<sup>1</sup><https://www.kaggle.com/>

<sup>2</sup><https://universe.roboflow.com/>

Table 1: Quantitative evidence for our third contribution. We report counting hallucination rates (CHR; the number of counting hallucinations divided by total generated samples) under different solver configurations and initial noise settings across the three datasets ToyShape, SimObject, and RealHand. For RealHand, we also report the non-counting failure rate (NCFR), which measures failure cases captured by the counting-ready indicator, and the total failure rate (TFR), defined as the sum of CHR and NCFR. *Diffused* and *Normal* refer to the ground-truth initial noise and standard Gaussian noise, respectively.

| Solver Name                      | Sampling Steps | CHR (ToyShape)  |               | CHR (SimObject) |               | CHR (RealHand)  |               | NCFR (RealHand) |               | TFR (RealHand)  |               |
|----------------------------------|----------------|-----------------|---------------|-----------------|---------------|-----------------|---------------|-----------------|---------------|-----------------|---------------|
|                                  |                | <i>Diffused</i> | <i>Normal</i> | <i>Diffused</i> | <i>Normal</i> | <i>Diffused</i> | <i>Normal</i> | <i>Diffused</i> | <i>Normal</i> | <i>Diffused</i> | <i>Normal</i> |
| DPM-Solver-1 (Song et al., 2021) | 25             | 2.43            | 3.47          | 9.27            | 9.95          | 12.71           | 12.95         | 16.18           | 18.06         | 28.90           | 31.01         |
|                                  | 50             | 1.83            | 2.79          | 8.53            | 9.27          | 13.82           | 13.85         | 11.32           | 12.51         | 25.14           | 26.36         |
|                                  | 100            | 1.56            | 2.32          | 8.04            | 8.90          | 14.29           | 14.55         | 9.54            | 10.63         | 23.83           | 25.19         |
| DPM-Solver-2 (Lu et al., 2022)   | 25             | 2.45            | 3.81          | 8.16            | 8.19          | 14.23           | 14.48         | 8.37            | 9.33          | 22.60           | 23.82         |
|                                  | 50             | 1.76            | 2.86          | 7.90            | 7.95          | 16.15           | 15.99         | 6.38            | 7.22          | 22.53           | 23.21         |
|                                  | 100            | 1.41            | 2.24          | 7.89            | 7.89          | 15.06           | 15.43         | 7.82            | 8.94          | 22.88           | 24.38         |
| DDPM (Ho et al., 2020)           | 1000           | 0.63            | 0.64          | 4.98            | 4.99          | 10.82           | 10.75         | 2.07            | 2.39          | 12.88           | 13.14         |

## 5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

This subsection delivers our third contribution: a systematic study of how denoising conditions shape counting hallucinations. Table 1 reports the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under varying denoising conditions, including solver types, ODE solver orders, and initial noise settings, across the three datasets ToyShape, SimObject, and RealHand. We distill the results into five observations, each of which we support directly from the table.

**Observation 1: Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand.** As shown in Table 1, under commonly used ODE solver settings (25, 50, and 100 steps), increasing the number of sampling steps generally decreases the counting hallucination rate on ToyShape and SimObject, yet tends to increase it on RealHand, except at 100 steps with DPM-Solver-2. This contrast suggests that synthetic datasets benefit from finer solver granularity due to their simpler, more structured distributions, whereas real-world datasets such as RealHand have more complex distributions in which additional steps may overfit local inconsistencies and thereby amplify hallucinations. This is the first piece of evidence for our headline finding that standard numerical strategies do not transfer to real objects.

**Observation 2: Higher-order ODE solvers mitigate non-counting failures but exacerbate counting hallucinations in RealHand under standard inference steps.** As shown in Table 1, comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling steps (e.g., 25 and 50) reveals a consistent trade-off: the higher-order solver substantially reduces non-counting failures while simultaneously raising the counting hallucination rate. A similar trend emerges when increasing the number of sampling steps for both solvers. The takeaway reinforces Observation 1: allocating more computation (i.e., “better” sampling conditions) reliably improves perceptual quality (lower non-counting failure rates) but does not enforce the factual correctness of finger counts.

**Observation 3: DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across all comparisons, ancestral sampling (i.e., DDPM) consistently achieves the lowest CHR, NCFR, and TFR, indicating that DDPM provides a practical lower bound for the failure rates of these generative models. This positions ancestral sampling as an effective, if computationally heavy, strategy for minimizing hallucinations.

**Observation 4: A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) tends to yield lower CHR, NCFR, and TFR than “normal” noise (i.e., standard Gaussian noise). This effect likely arises because diffused noise better matches the initial noise distribution seen during training, indicating that a more informed initialization can mitigate not only counting hallucinations but also other non-counting failures.

**Observation 5: Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that

Table 2: Quantitative backing for our fourth contribution: the correlation of counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with the perceptual metrics FID and MUSIQ. The weak CHR correlations versus the strong NCFR correlations are the key contrast to read off this table. Results for PTI-Hand are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here,  $p$  denotes the p-value assessing the statistical significance of the correlation  $r$  or  $\rho$ .  $\dagger$  means DDPM results integrated with those from various ODE solvers across sampling conditions.

| Metric                                    | CHR vs. FID |       |          |       |          |      | NCFR vs. FID |       |          |       |  |  | CHR vs. MUSIQ |       |          |      |  |  | NCFR vs. MUSIQ |       |          |       |  |  |
|-------------------------------------------|-------------|-------|----------|-------|----------|------|--------------|-------|----------|-------|--|--|---------------|-------|----------|------|--|--|----------------|-------|----------|-------|--|--|
|                                           | SimObject   |       | RealHand |       | PTI-Hand |      | RealHand     |       | PTI-Hand |       |  |  | RealHand      |       | PTI-Hand |      |  |  | RealHand       |       | PTI-Hand |       |  |  |
|                                           | $r$         | $p$   | $r$      | $p$   | $r$      | $p$  | $r$          | $p$   | $r$      | $p$   |  |  | $r$           | $p$   | $r$      | $p$  |  |  | $r$            | $p$   | $r$      | $p$   |  |  |
| Pearson ( $r$ )                           | 0.87        | 0.01  | -0.97    | <0.01 | 0.20     | 0.69 | 0.99         | <0.01 | 0.88     | <0.01 |  |  | 0.93          | <0.01 | -0.02    | 0.95 |  |  | -0.97          | <0.01 | -0.99    | <0.01 |  |  |
| Pearson $^\dagger$ ( $r$ )                | 0.49        | 0.14  | -0.53    | 0.13  | —        | —    | 0.96         | <0.01 | —        | —     |  |  | 0.46          | 0.20  | —        | —    |  |  | -0.98          | <0.01 | —        | —     |  |  |
| Spearman ( $\rho$ )                       | 0.84        | <0.01 | -0.97    | <0.01 | -0.14    | 0.78 | 0.99         | <0.01 | 0.92     | <0.01 |  |  | 0.80          | 0.01  | 0.42     | 0.33 |  |  | -0.88          | <0.01 | -0.46    | 0.29  |  |  |
| Spearman $^\dagger$ ( $\rho$ )            | 0.75        | 0.01  | -0.51    | 0.15  | —        | —    | 0.95         | <0.01 | —        | —     |  |  | 0.40          | 0.28  | —        | —    |  |  | -0.91          | <0.01 | —        | —     |  |  |
| Distance Correlation ( $r_d$ )            | 0.89        | —     | 0.97     | —     | 0.47     | —    | 0.99         | —     | 0.91     | —     |  |  | 0.93          | —     | 0.42     | —    |  |  | 0.97           | —     | 0.99     | —     |  |  |
| Distance Correlation $^\dagger$ ( $r_d$ ) | 0.82        | —     | 0.74     | —     | —        | —    | 0.95         | —     | —        | —     |  |  | 0.71          | —     | —        | —    |  |  | 0.97           | —     | —        | —     |  |  |

diffusion models exhibit higher CHR as the morphological complexity of the target objects increases. This indicates that the fine-grained details and complex spatial relationships of morphologically complex objects make accurate object counts harder to maintain, leading to more frequent additions or omissions; we exploit exactly this observation later to motivate our mitigation paradigm in Sec. 5.5.

### 5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

This subsection presents our fourth contribution and answers a question at the heart of it. Counting hallucination is a form of factualness hallucination, so it is natural to ask: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our answer, supported by two pieces of evidence below, is **not necessarily**.

To examine this relationship, we perform correlation analysis between CHR and perceptual metrics, and between NCFR and the same metrics. We focus on two widely used perceptual measures: a distribution-based metric, FID (Heusel et al., 2017), and an instance-based metric, MUSIQ (Ke et al., 2021). For correlation estimation, we adopt three complementary measures: Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear dependence), Spearman rank correlation (Spearman, 1904) (monotonic dependence), and Distance Correlation (Székely et al., 2007) (general nonlinear dependence).

🔑 **Evidence 1: The correlations between CHR and perceptual metrics are dataset- and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, the Pearson correlations between CHR and FID vary substantially across datasets (positive for SimObject, negative for RealHand, and positive again for PTI-Hand), while the corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also fall from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that more complex generative content weakens the association between CHR and FID. Integrating DDPM results with those from ODE solvers attenuates these correlations further still.

🔑 **Evidence 2: The correlations between NCFR and perceptual metrics are consistently strong and positive.** In sharp contrast to the weak CHR correlations, the correlation between NCFR and perceptual metrics is strong and robust across all settings. For example, all correlation coefficients exceed 0.88 for NCFR vs. FID, with very small p-values, indicating high statistical significance. Taken together, the two pieces of evidence establish our fourth contribution: perceptual metrics track perceptual failures but not counting hallucinations.

### 5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

A natural concern is whether findings obtained on our controlled datasets and the accompanying protocol extend beyond them, so we test both on a large-scale pretrained text-to-image model.

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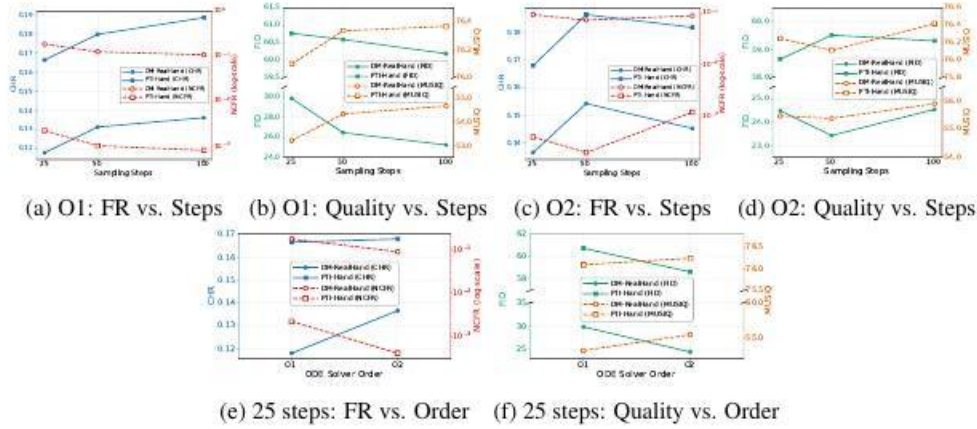


Figure 3: Evidence that our third-contribution findings generalize: trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCFR; quality metrics include FID and MUSIQ. Across variations in sampling steps and solver order, both models exhibit highly consistent trends, confirming that the conclusions transfer to a large pretrained model.

Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model introduced in Sec. 4.2.3, with further details in Appendix A.8. As shown in Figure 3, the way solver type and sampling steps affect counting hallucinations in this large pretrained model aligns closely with the trends from our controlled RealHand experiments. This consistency both validates the findings of Sec. 5.2 (our third contribution) and shows that our evaluation protocol (our second contribution) applies well beyond CountHalluSet. The correlation results in Table 2 tell the same story: we again observe weak correlations between CHR and perceptual metrics and strong correlations between NCFR and perceptual metrics. Together, these results demonstrate the robustness and generality of our findings and confirm that the proposed protocol transfers reliably to pretrained text-to-image models.

## 5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

Our final contribution turns the diagnosis into a remedy. The lower counting hallucination rates observed for simpler countable objects (Observation 5 in Sec. 5.2) suggest that diffusion models capture counting-based factuality more reliably when the underlying visual structure has low morphological complexity (e.g., masked objects). This directly motivates the question we answer here: if a diffusion model denoised general visual representations *together with* explicit, simple structural constraints that are easier to model, such as object masks that encode basic topology and counts, within a shared latent space, could such a joint learning paradigm mitigate counting hallucinations? We refer to models that jointly learn general visual representations and structural constraints during denoising as **joint-diffusion models (JDM)**.

Concretely, we perform a channel-wise concatenation of the latent representation of the hand mask with that of the corresponding RGB image, as shown in Fig. 4. We then finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration in Sec. 5.1) using the RealHand images and their associated masks, extracted with SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e.,  $3 + 3 = 6$  channels).

The results validate the paradigm. As shown in Table 3, JDM consistently achieves lower failure rates than the vanilla LDM on both CHR and NCFR across diverse sampling conditions, demonstrating its effectiveness at accurately generating structurally intricate objects that are prone to counting ambiguities. The one cost is a higher FID, which likely stems from the increased denoising burden of jointly modeling two modalities under a small-scale training regime, trading some perceptual quality for improved structural realism. Encouragingly, we also observe promising reductions in

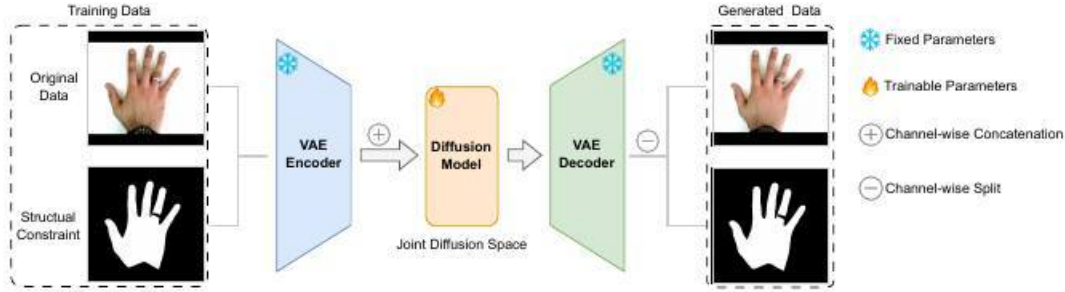


Figure 4: Our fourth contribution at a glance: the proposed joint-diffusion model (JDM). The model performs channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. This joint learning paradigm encourages anatomically plausible hand poses and correct finger topology during generation, which is what yields the reduced CHR and NCFR.

Table 3: Evidence for our mitigation contribution: failure rates of the vanilla LDM versus our proposed JDM on the RealHand dataset across different ODE solvers and sampling steps. Lower values indicate better performance; the consistent CHR and NCFR reductions are the result to read off this table. For JDM, although the model jointly generates hand RGB images and segmentation masks, FID is computed solely on the RGB outputs to ensure comparability with the baseline.

| Solver Name    | Counting Hallucination Rate (CHR) ↓ |       |       |              |       |       | Non-counting Failure Rate (NCFR) ↓ |       |       |              |      |      | Fréchet Inception Distance (FID) ↓ |       |       |              |       |       |
|----------------|-------------------------------------|-------|-------|--------------|-------|-------|------------------------------------|-------|-------|--------------|------|------|------------------------------------|-------|-------|--------------|-------|-------|
|                | DPM-Solver-1                        |       |       | DPM-Solver-2 |       |       | DPM-Solver-1                       |       |       | DPM-Solver-2 |      |      | DPM-Solver-1                       |       |       | DPM-Solver-2 |       |       |
| Sampling Steps | 25                                  | 50    | 100   | 25           | 50    | 100   | 25                                 | 50    | 100   | 25           | 50   | 100  | 25                                 | 50    | 100   | 25           | 50    | 100   |
| LDM            | 12.95                               | 13.85 | 14.55 | 14.48        | 15.99 | 15.43 | 18.06                              | 12.51 | 10.63 | 9.33         | 7.22 | 8.94 | 29.86                              | 26.43 | 25.21 | 24.48        | 23.42 | 24.54 |
| JDM (ours)     | 11.41                               | 10.67 | 10.51 | 10.12        | 9.66  | 10.07 | 2.94                               | 2.42  | 2.29  | 3.51         | 4.63 | 2.26 | 37.69                              | 35.51 | 34.68 | 34.63        | 32.07 | 33.89 |

counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings, with more details, results, and discussion in Appendix A.9.

## 6 CONCLUSION

This work takes the first step toward systematically quantifying hallucinations in diffusion models, and its contributions map directly onto the four we set out in the introduction. *First*, we defined counting hallucination as a specific, measurable form of factual error. *Second*, we introduced **CountHalluSet** together with a practical and reproducible methodology for measuring counting hallucinations. *Third*, through comprehensive and controlled experiments we showed that commonly adopted numerical strategies often fail to mitigate counting hallucinations when generating human hands. *Fourth*, our correlation analyses demonstrated that widely used perceptual metrics (e.g., FID and MUSIQ) cannot consistently reflect the severity of counting hallucinations, and, building on these findings, our joint-diffusion paradigm reduced both counting hallucinations and non-counting failures on RealHand. Experiments on large-scale pretrained text-to-image models further corroborate both the generality and the validity of these findings, and we hope counting hallucination serves as a tractable entry point for the broader goal of factually consistent generation.

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## A APPENDIX

### A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process  $q(\mathbf{x}_{T:0})$  and the true reverse process  $p_\theta(\mathbf{x}_{T:0})$ , their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[ \log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes  $q(\mathbf{x}_{T:0})$  and  $p_\theta(\mathbf{x}_{T:0})$ , we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[ \log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[ \log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[ \log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[ \mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[ \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality (=) would strictly hold only in the ideal case where  $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$  and  $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$  for all  $t$  and  $\mathbf{x}_t$ , which means the diffusion model can accurately recover the underlying data distribution  $q(\mathbf{x}_0)$ . This upper bound for the overall KL divergence ( $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ ) between true data and model-generated data is identical to the variational bound derived in DDPM (Ho et al., 2020) when data entropy  $H(\mathbf{x}_0)$  is excluded, thereby suggesting that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between trued data and model-generated data.

**Diffused-prior gap.** The first source of errors arises from the mismatch between the distribution of the ground-truth diffused data  $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$  and the prior distribution used for sampling  $p_\theta(\mathbf{x}_T)$ . This mismatch stems from the fact that the number of diffusion steps  $T$  is finite in practice. The discrepancy can be formally quantified using a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where  $\mathcal{D}_{\text{KL}}$  denotes the Kullback–Leibler (KL) divergence. Alternatively, other appropriate metrics such as FID or the Wasserstein distance can be employed to evaluate this distributional discrepancy. The diffused-prior gap, present from the outset, induces a deviation that persists throughout the denoising process, ultimately resulting in a sustained output error.

**Noise model prediction error.** The second source of errors stems from the noise prediction model. Because the model’s estimate  $\epsilon_\theta(\mathbf{x}_t, t)$  only approximates the true noise  $\epsilon$ , as guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let  $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$  represent this prediction error. Based on Eq. (5), any discrepancy in  $\epsilon_\theta$  induces a corresponding deviation in the predicted mean at step  $t$ :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

## A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver approximates the solution as example. Let  $\mathbf{x}^*(\tau)$  denote the ideal trajectory that satisfies this ODE with the true score function  $h^*(\mathbf{x}(\tau), \tau)$  and starts from a ground-truth initial condition  $\mathbf{x}^*(T)$  drawn from the true diffused distribution  $q_T(\mathbf{x}_T)$ . Similarly,  $\hat{\mathbf{x}}_k$  represent the numerical trajectory computed by a numerical ODE solver at discrete time steps  $\tau_k = T - k\Delta t$  for  $k = 0, 1, \dots, N$  and  $\hat{h}$  is the estimated model function. For a specific numerical trajectory from  $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$ , we want to calculate its total accumulated trajectory error. Let  $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$  be the trajectory error at step  $k$ . Consider the transition from  $\tau_k$  to  $\tau_{k+1} = \tau_k - \Delta t$ .

**Ideal path evolution.** Using the variation-of-constants formula for the exact solution over one step (from  $\tau_k$  down to  $\tau_{k+1}$ ):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where  $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$  is the state transition operator for the linear part  $\frac{dz}{d\tau} = f(\tau)z$ . Note that the integration limits are backward in time.

**Numerical path evolution.** A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by  $\hat{h}$  starting from  $\hat{\mathbf{x}}_k$ . Let  $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$  be the exact solution at  $\tau_{k+1}$  of the ODE using  $\hat{h}$ , starting from  $\hat{\mathbf{x}}_k$  at  $\tau_k$ :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error  $\delta_k$ :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

**Error recurrence.** Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[ \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

**Accumulated error formulation.** We can unroll this recurrence relation from  $k = 0$  to  $N - 1$ . Let  $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$  for  $j > k$ , and  $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$ . Applying the recurrence

repeatedly:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left( \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[ \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting  $\tau_N = 0$ ,  $\tau_0 = T$ ,  $e_N = \mathbf{x}_N - \mathbf{x}^*(0)$ , and  $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ :

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left( \underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[ \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term represents the deterministic impact on the final path error of a specific initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

### A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with  $\eta = 0$ ) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \left( \frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward (diffusion) process, DDIM does not simply strip the stochastic term from DDPM's reverse (denoising) process. This is because the two methods employ different coefficient formulations for the predicted noise term.

### A.4 DATASET EXAMPLES AND DETAILS

This appendix documents the CountHalluSet suite behind our second contribution in more detail. Figure A1 presents representative examples from the three datasets used in this study. As shown in Table A1, each dataset has a balanced distribution across its categories, and Table A2 further shows that the distribution of images with different object counts is comparable, confirming the absence of object-count bias across datasets. The datasets appear in the same order in both tables, and the category order in Table A1 is as follows:

- **ToyShape**: triangle, square, pentagon;
- **SimObject**: mug, apple, clock;
- **RealHand**: finger.

Table A1: Category balance in CountHalluSet: number of objects per category for each dataset.

|            | ToyShape | SimObject | RealHand |
|------------|----------|-----------|----------|
| Category 1 | 20,027   | 19,916    | 25,250   |
| Category 2 | 19,902   | 20,101    | -        |
| Category 3 | 19,978   | 19,974    | -        |

Table A2: Object-count balance in CountHalluSet: number of images by object count(s) per dataset.

|           | ToyShape | SimObject | RealHand |
|-----------|----------|-----------|----------|
| 1 object  | 10,003   | 10,022    | -        |
| 2 objects | 10,087   | 9,965     | -        |
| 3 objects | 9,910    | 10,013    | -        |
| 5 objects | -        | -         | 25,250   |
| Total     | 30,000   | 30,000    | 25,250   |

#### A.5 EXAMPLES OF COUNTING HALLUCINATIONS

To make the definition behind our first contribution concrete, we provide representative examples of counting hallucinations for all three datasets in Fig. A2. For ToyShape and SimObject, counting hallucinations either contain more than two instances of at least one category or contain no instances at all against a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples are characterized by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear to display five fingers, yet one finger is partially incomplete and visibly artifacted, effectively leaving four valid fingers; these samples are therefore classified as counting hallucinations.

#### A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

To document the reliability of the counting model that underpins our protocol, Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector accurately identifies fingertips across diverse hand poses and orientations, including both dorsal and palmar views. It may nonetheless fail on anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and thus remain indistinguishable from anatomically valid ones in terms of finger counts.

Figure A3(b) shows several counting-hallucinated samples where YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips), and Figure A3(c) illustrates several rare edge cases (<1%) in which the detector fails on fingertips with unclear boundaries, cracked fingers, or floating artifacts. Such rare cases can be partly mitigated by tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds), but doing so may compromise detection performance on the majority of counting-valid samples. Because a consistent detection setup is applied across all RealHand evaluations, these rare cases do not affect the validity of our overall conclusions, even though fully addressing them remains a challenging and low-reward effort.

#### A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

To clarify the two-stage RealHand protocol at the core of our second contribution, Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the RealHand evaluation procedure first applies the counting-ready indicator to separate counting-ready samples (i.e., counting-valid and counting-hallucinated) from images unsuitable for counting, the latter typically exhibiting severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. Counting-ready

samples are then classified as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

## A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

These details support the generalization study of Sec. 5.4. In the PTI-Hand experiments, we prompt Stable-Diffusion-3.5-Medium with 10 different hand-generation prompts, listed as follows.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Our definition of hallucination assumes the model generates patterns never observed during training, an assumption that is difficult to verify for large-scale pretrained diffusion models whose full training distribution is inaccessible. To keep the quantification of counting hallucinations feasible in this setting, we adopt a conservative criterion: any generated hand image containing six or more fingers is treated as a counting-hallucinated sample. This is well justified, because while real images may depict fewer than five visible fingers due to occlusion, articulation, or viewpoint, real data almost never contains hands with six or more fingers. Finger counts  $\geq 6$  therefore provide a reliable and empirically verifiable signal of counting hallucination, so that our protocol remains applicable even without access to the training distribution; generated examples appear in Fig. A5.

## A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

### A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

This appendix expands on the joint-diffusion paradigm that constitutes our mitigation contribution. In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared to 80k steps for the vanilla LDM. This larger training budget is necessary because extending the latent space from three to six channels in JDM introduces a substantial mismatch with the pretrained latent distribution of LDM. Additional optimization is therefore required to close this distributional gap and let JDM’s modeling capacity fully materialize.

### A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

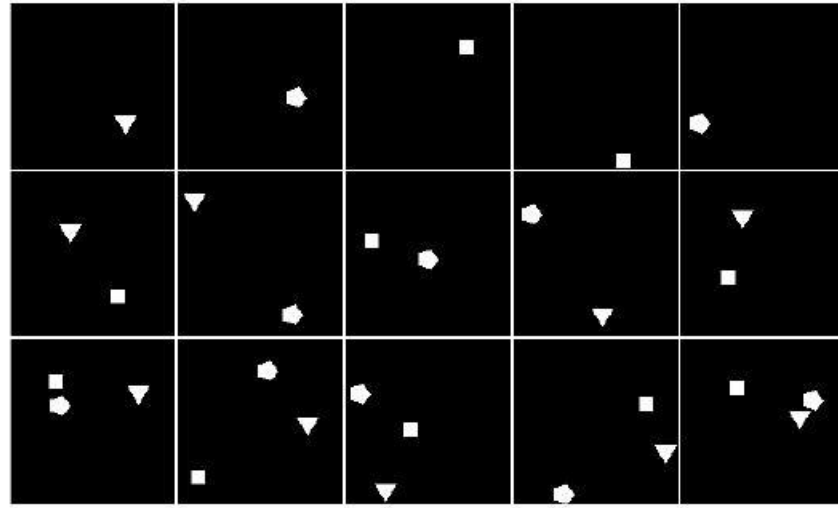
We stress that JDM should be understood as a **novel training paradigm** for improving factual generation in diffusion models, not as a superior inference strategy, for two reasons. First, JDM only extends the input and output channels of the UNet while leaving all intermediate layer sizes unchanged, so it **introduces negligible inference-time overhead compared to the vanilla LDM**. Second, extending a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and undermines the original denoising distribution**, which makes directly finetuning such a model nontrivial.

Table A3: Extending our mitigation contribution to a pretrained model: failure rates and image quality when finetuning SD-1.5 versus finetuning SD-1.5 with the joint-diffusion training paradigm on RealHand. The first row (baseline) reports finetuning Stable-Diffusion-1.5 on RealHand data. The second row freezes all intermediate layers of the baseline model and finetunes only the input and output layers with extended channel dimensions. Lower values indicate better performance.

| Solver Name                            | Counting Hallucination Rate (CHR) ↓ |        |        |              |        |        | Non-counting Failure Rate (NCFR) ↓ |        |        |              |        |        | Fréchet Inception Distance (FID) ↓ |       |        |              |       |       |
|----------------------------------------|-------------------------------------|--------|--------|--------------|--------|--------|------------------------------------|--------|--------|--------------|--------|--------|------------------------------------|-------|--------|--------------|-------|-------|
|                                        | DPM-Solver-1                        |        |        | DPM-Solver-2 |        |        | DPM-Solver-1                       |        |        | DPM-Solver-2 |        |        | DPM-Solver-1                       |       |        | DPM-Solver-2 |       |       |
| Sampling Steps                         | 25                                  | 50     | 100    | 25           | 50     | 100    | 25                                 | 50     | 100    | 25           | 50     | 100    | 25                                 | 50    | 100    | 25           | 50    | 100   |
| SD-1.5                                 | 0.3362                              | 0.3404 | 0.3630 | 0.3396       | 0.3376 | 0.3362 | 0.0059                             | 0.0032 | 0.0020 | 0.0050       | 0.0026 | 0.0018 | 61.51                              | 57.71 | 57.70  | 58.53        | 57.57 | 56.69 |
| SD-1.5 with joint-diffusion finetuning | 0.2838                              | 0.3222 | 0.4145 | 0.3053       | 0.3277 | 0.3558 | 0.0947                             | 0.0511 | 0.0487 | 0.0549       | 0.0362 | 0.0327 | 70.65                              | 79.22 | 102.39 | 70.80        | 72.84 | 83.07 |

Even so, we still observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning conditions. In practice, aligning JDM’s extended latent space with the highly rigid latent geometry of large pretrained models proves challenging under the limited scale of RealHand. To strike a practical balance, we adopt a two-stage finetuning strategy: we first finetune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with extended channels. As shown in Table A3, even under this highly constrained setup the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence in this work supports JDM as a promising direction for enforcing factual consistency in image generation, and we believe that **studying its behavior and performance under true large-scale training remains an important and compelling avenue for future research.**



(a) Examples in the ToyShape dataset

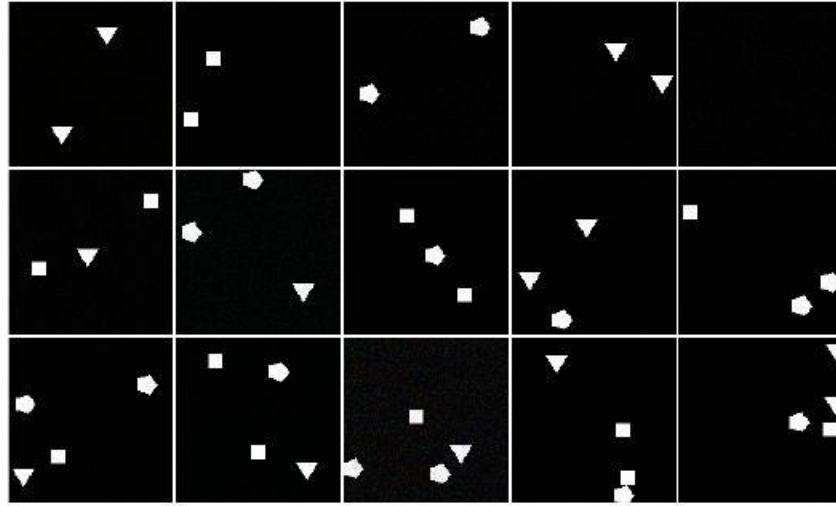


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

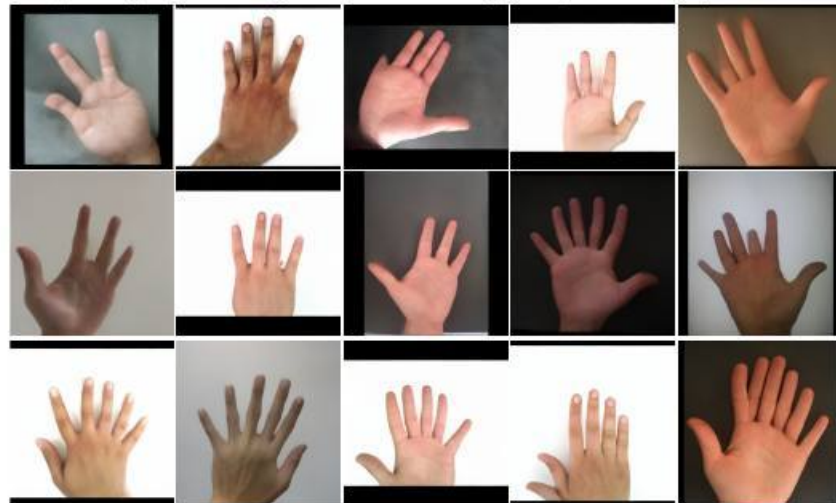
Figure A1: Examples of datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

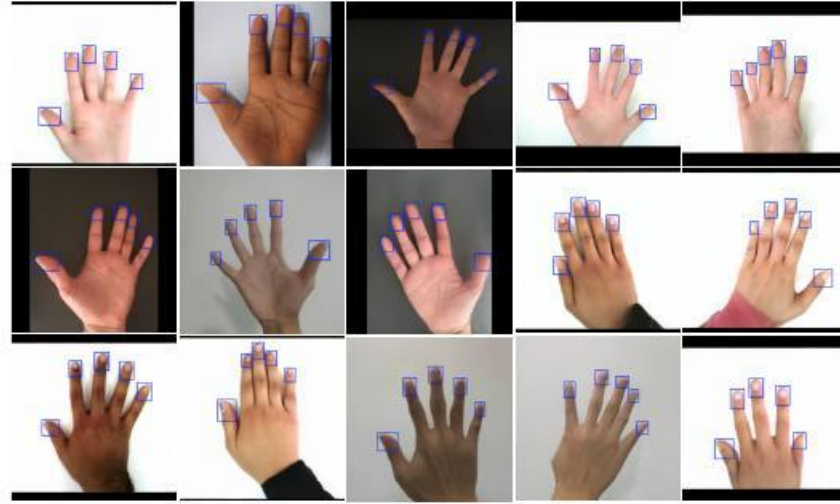


(b) Examples of counting hallucinations corresponding to the SimObject dataset

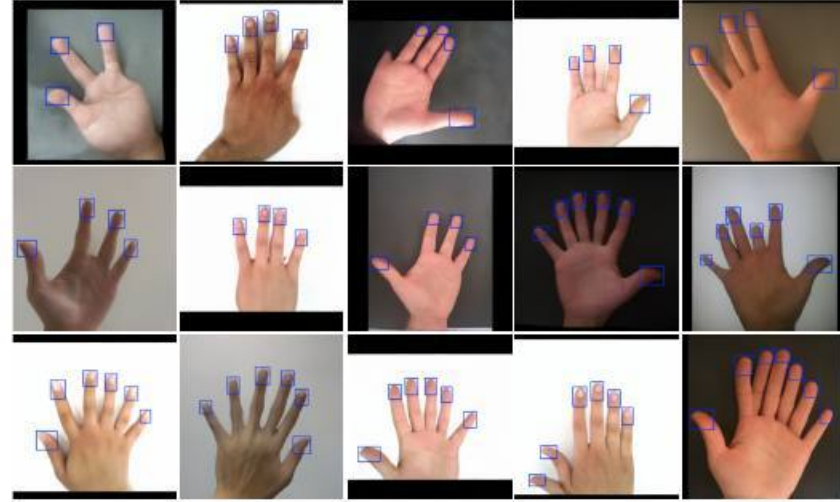


(c) Examples of counting hallucinations corresponding to the RealHand dataset

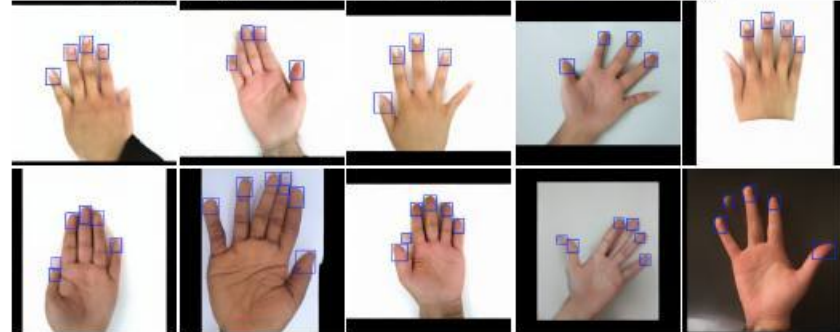
Figure A2: Examples of counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

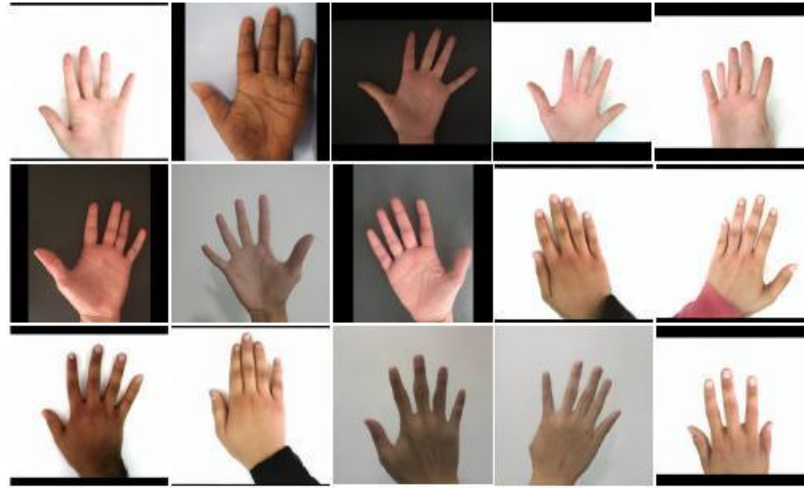


(b) Counting-hallucinated YOLO predictions with not five fingers

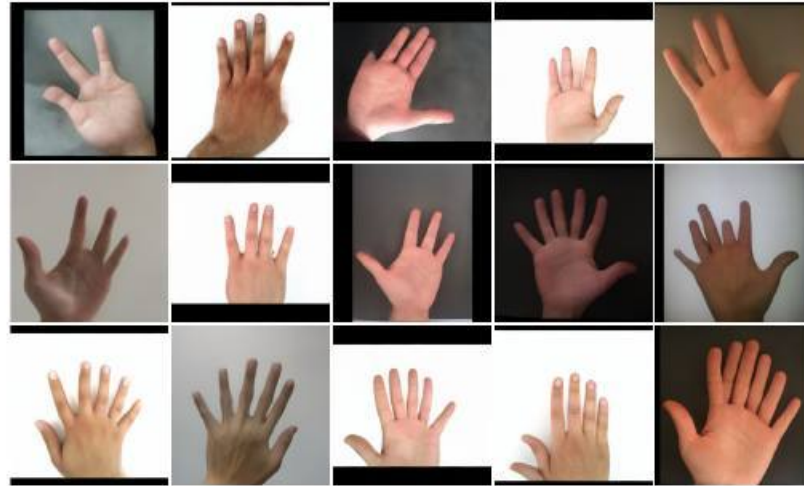


(c) Rare edge cases where YOLO predictions fail

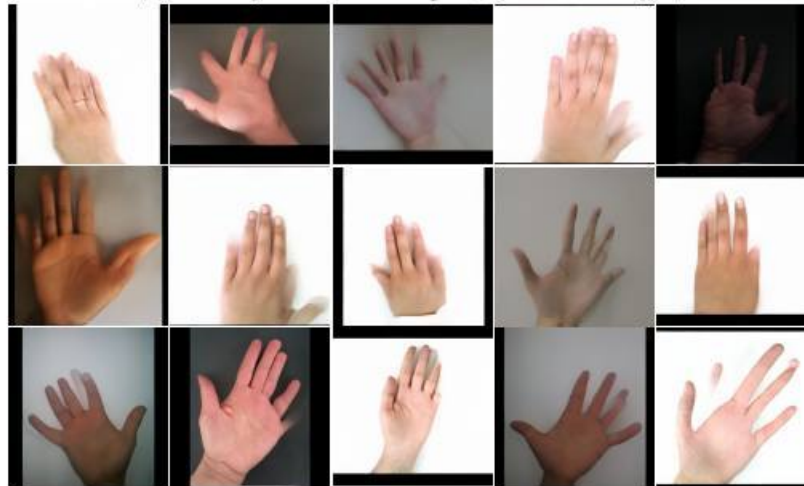
Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases where the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of three classes of generated hand images using pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.